CheckMATE implementation of CMS-NPS-025-002 (`cms_2605_13614`)

13 and 13.6 TeV analyses and the statistical combination with Combine

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1 Overview

The CMS search for pair production of additional neutral scalars of the Inert Doublet Model (IDM) in final states with two electrons or two muons and missing transverse momentum uses the full Run 2 (2016–2018, $\sqrt{s} = 13\,\text{TeV}$) and the 2022 (Run 3, $\sqrt{s} = 13.6\,\text{TeV}$) data sets. The signal is extracted with a parametrised neural network (pNN) whose output is fitted in the full CMS statistical model with Combine.

In CheckMATE the analysis is split into two implementations, one per centre-of-mass energy, and the results of the two CheckMATE runs are combined afterwards:

Analysis	$\sqrt{s}$	Data sets	Detector setup (experiment)
`cms_2605_13614`	13 TeV	2016, 2017, 2018	`cms13tev`
`cms_2605_13614_13p6`	13.6 TeV	2022	`cms13_6tev`

The full workflow is:

1. generate signal events at 13 and 13.6 TeV,
2. run CheckMATE on the 13 TeV events with the analysis `cms_2605_13614` and on the 13.6 TeV events with the analysis `cms_2605_13614_13p6`,
3. run `bin/combine_cms_2605_13614.py` on the two runs to obtain the limits for each IDM mass point.

2 Requirements

- CheckMATE compiled with ONNX Runtime support (`HAVE_ONNX`); the pNN model is read from `data/cms_2605_13614/pnn_model_with_preprocessing.onnx`.
- The CMS datacards in `data/cms_2605_13614/datacards`:
  - the main datacard `input/combined_cards.txt`,
  - the histograms of every mass point, `input/histograms/mH<mH>_mA<mA>/histograms.root` (about 17 MB per mass point, 9 GB in total).
- PyROOT, used to update the signal histograms.
- Combine available in the `PATH` (the CMS results were produced with version v10.5.0). It may be necessary to run `ulimit -s unlimited; ulimit -u unlimited` before running Combine.

3 The CMS 13.6 TeV detector setup

The analysis handler `AnalysisHandlerCMS_13_6TeV` (`tools/fritz/src/analysishandler`) provides the detector setup for CMS analyses at $\sqrt{s} = 13.6$ TeV. All its settings, i.e. the photon, electron and muon identification efficiencies and the b- and τ-tagging efficiencies, are taken over unchanged from the 13 TeV handler `AnalysisHandlerCMS_13TeV`. The Delphes card of the 13 TeV setup, `delphes_skimmed_CMS_13TeV.tcl`, is used as well.

The setup is registered in CheckMATE as

Experiment	`cms13_6tev`
Analysis group	`CMS_13_6TeV`
Centre-of-mass energy	13.6 TeV
Analysis list	`data/analysis_info/CMS_13_6TeV_analyses.txt`
Analysis sources	`tools/analysis/{src,include}/analyses/CMS_13_6TeV`

New 13.6 TeV analyses are added with the `AnalysisManager` by choosing CMS and a centre-of-mass energy of 13.6 TeV. The experiment recorded in the analysis settings file (`_var.j`) selects the analysis handler at run time:

```
"ecm": "13.6",
"experiment": "cms13_6tev",
```

Note. Analysis names must be unique in CheckMATE across all analysis groups. A 13.6 TeV version of an existing analysis therefore needs a new name, e.g. with the suffix `_13p6`. When the settings file of an existing analysis is copied by hand, the `experiment` entry has to be changed as well; otherwise the analysis is booked in the 13 TeV handler and CheckMATE stops with *"Cannot load analysis ... – analysis unknown"*.

4 The analyses

4.1 Event selection and pNN evaluation

Both implementations apply the event preselection of the CMS analysis to events with two same-flavour leptons and evaluate the pNN for every mass point (m_H, m_A) of the mass grid (Section 4.3). Events with a pNN output of at least 0.9 are counted in the bins of the pNN distribution used by CMS: 7 bins in the electron channel and 11 bins in the muon channel.

- `cms_2605_13614` (13 TeV, $\mathcal{L} = 138\,\text{fb}^{-1}$): every event is assigned at random to one of the data-taking years 2016, 2017 and 2018, with probabilities 25.3 %, 27.8 % and 46.9 % according to the luminosities of the years. The year-dependent lepton trigger thresholds are applied.
- `cms_2605_13614_13p6` (13.6 TeV, $\mathcal{L} = 35\,\text{fb}^{-1}$): all events are assigned to 2022.

4.2 Output: control regions

The yields are stored as control regions in the CheckMATE output file

`results/<run>/analysis/<process>_<analysis>_control.dat`

with one control region per mass point, year, channel and pNN bin:

`mH<mH>_mA<mA>_<year>_<channel>_bin_<n>`

where `<channel>` is `EE` (bins `0`–`6`) or `MM` (bins `00`–`10`), e.g. `mH70.0_mA140.0_2016_EE_bin_3`. Both analyses book the regions of all four years, so that the output files of the two energies have the same structure; the regions of the years not covered by an analysis remain empty.

4.3 Mass grid

The mass points are defined in the `mHmA` list in the analysis headers `cms_2605_13614.h` and `cms_2605_13614_13p6.h`. The list contains all 524 mass points for which CMS provides histograms in the datacards:

- $m_H = 60$–$150\,\text{GeV}$ in steps of $2.5\,\text{GeV}$,
- m_A mostly on a $5\,\text{GeV}$ grid with $20\,\text{GeV} \le m_A - m_H \le 100\,\text{GeV}$, including additional points at $m_A = m_H + 92\,\text{GeV}$; for $m_H \ge 135\,\text{GeV}$ the grid is $m_A = m_H + 20, 30, \ldots, 100\,\text{GeV}$ and $m_H + 92\,\text{GeV}$.

The exact list of points is given in the headers. Since the control regions are cheap to produce, CheckMATE always evaluates the full grid. The selection of the mass points for the statistical evaluation is done in the combination script.

5 Running CheckMATE

The two energies are run separately, each with events generated at the corresponding centre-of-mass energy. Example parameter files:

```
[Parameters]
Name: wino_320_40_decay
Analyses: cms_2605_13614

[hhh]
Events: /path/to/events_13TeV.hepmc
XSect: 18.8 fb
```

```
[Parameters]
Name: wino_320_40_decay_136
Analyses: cms_2605_13614_13p6

[hhh]
Events: /path/to/events_13p6TeV.hepmc
XSect: 20.8 fb
```

All control-region files of a run (one per process) are taken into account in the combination.

6 Statistical combination: `combine_cms_2605_13614.py`

6.1 Synopsis

```
bin/combine_cms_2605_13614.py run [run ...]
    [--mH MH --mA MA | --all]
    [--datacards DIR] [--output DIR]
    [--r-max R] [--combine-executable EXE]
    [--prepare-only] [--keep-histograms]
```

The script is generated by `configure` from `tools/python/combine_cms_2605_13614.in`, in the same way as `bin/CheckMATE`. It therefore uses the same results and data folders as CheckMATE: the results folder is `results/` in the CheckMATE directory, or the folder given in the environment variable `CHECKMATE_RESULTS`.

6.2 Description

The script

1. collects the files `*_cms_2605_13614*_control.dat` from `<results>/<run>/analysis/` of all given runs, i.e. the output of both `cms_2605_13614` and `cms_2605_13614_13p6`;
2. sums the yields of every control region over all files. The statistical uncertainty of each file, $\Delta N = N/\sqrt{A \cdot N_{\mathrm{MC}}}$ with the acceptance A and the number of generated events N_{MC}, is added in quadrature. The result is written to `cms_2605_13614_combined_control.txt` in the output folder;
3. evaluates the selected mass points one after another. For every mass point it
   (a) copies the original `histograms.root` of the mass point from the datacards to the output folder,
   (b) replaces the signal histograms of the copy (the `nominal` histogram and all systematic variations in `<year>/SR/{Electron,Muon}/idm`) by the combined yields,
   (c) runs
   ```
   combine combined_cards.txt -M AsymptoticLimits \
       --keyword-value MA=<mA> --keyword-value MH=<mH> \
       --strictBounds --rMax <R> -n _mH<mH>_mA<mA>
   ```
   in the output folder, which contains a copy of `combined_cards.txt`,
   (d) extracts the observed and expected limits on the signal strength r and adds them to `limits.txt`,
   (e) deletes the modified histogram file.

Handling of the histograms. The datacards are never modified. Only one modified histogram file (about 17 MB) exists at any time, and it is deleted after the mass point has been evaluated, also when the evaluation fails or the script is interrupted (Ctrl-C). Modified histograms are only kept with `--keep-histograms` or `--prepare-only`; note that this requires about 1.5 GB for the default selection and 9 GB with `--all`.

6.3 Arguments and options

`run [run ...]`	Names of the CheckMATE runs to combine (folders in the results folder), typically the 13 TeV and the 13.6 TeV run. A single run can be given as well.
`--mH MH --mA MA`	Evaluate a single mass point. Both masses have to be given, and the point has to be in the mass grid of the analysis and have histograms in the datacards.
`--all`	Evaluate all 524 mass points of the analysis.
`--datacards DIR`	Folder with the Combine datacards; default: `data/cms_2605_13614/datacards`.
`--output DIR`	Output folder (default: `combined_<run1>_<run2>...` in the results folder).
`--r-max R`	Maximum signal strength passed to Combine (default: 10).
`--combine-executable EXE`	Combine executable (default: `combine`).
`--prepare-only`	Only prepare the histograms, do not run Combine. The modified histograms are kept.
`--keep-histograms`	Keep the modified histograms after running Combine.

6.4 Selection of mass points

Command line	Evaluated mass points
(default)	90 points with m_H and m_A on a 10 GeV grid
`--all`	all 524 points
`--mH X --mA Y`	the single point (X, Y), any of the 524 points

`--all` cannot be combined with `--mH`/`--mA`. Mass points without histograms in the datacards are skipped with a warning.

6.5 Output

The output folder contains

`cms_2605_13614_combined_control.txt` combined control-region yields and uncertainties of all mass points,

`combined_cards.txt` copy of the Combine datacard,

`limits.txt` table of the limits on r per mass point (see below),

`logs/mH<mH>_mA<mA>.log` full Combine output of every mass point,

`higgsCombine_mH<mH>_mA<mA>.AsymptoticLimits.mH120.root`
Combine output file of every mass point.

`limits.txt` has the columns

```
mH  mA  obs  exp2.5  exp16  exp50  exp84  exp97.5  status
```

with the observed limit, the expected limit and its quantiles, and the status `ok`, `failed` or `prepared` (`--prepare-only`); missing values are written as `-`. A mass point is excluded at 95 % CL if $r_{\mathrm{obs}} < 1$. Results of earlier calls with the same output folder are kept, so the table can be filled point by point. The script ends with exit code 1 if any mass point failed and lists these points.

6.6 Examples

Evaluate a single mass point:

```
bin/combine_cms_2605_13614.py wino_320_40_decay wino_320_40_decay_136 \
    --mH 70 --mA 140
```

Evaluate the default selection of 90 mass points:

```
bin/combine_cms_2605_13614.py wino_320_40_decay wino_320_40_decay_136
```

Evaluate all mass points:

```
bin/combine_cms_2605_13614.py wino_320_40_decay wino_320_40_decay_136 --all
```

Prepare the histograms of one mass point for inspection without running COMBINE:

```
bin/combine_cms_2605_13614.py wino_320_40_decay wino_320_40_decay_136 \
    --mH 70 --mA 140 --prepare-only --output /tmp/check
```

6.7 Helper scripts

The script uses two modules in `tools/python` that can also be run on their own:

`combine_control.py <folder> <analysis>`
: sums the control-region files `*_<analysis>_control.dat` in one folder and writes the result to `<analysis>_combined_control.txt`.

`combine_wrapper.py <mH> <mA> --control-file F --histogram-file H`
: fills the combined yields of one mass point into a histogram file. *This modifies the given file in place.*